

Review Results

Contribution Details

Submission Type / Conference Track: CDIO Implementation Paper

USING PULL REQUESTS TO MAKE COLLABORATION VISIBLE IN CDIO PROJECT-BASED COURSES

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Topics: Industrial collaborations that enhance student experience and education, Curriculum development, Innovative learning environments

Keywords: Pull requests, GitHub Classroom, assessment infrastructure, team-based learning, professional competence, CDIO standards: 5, 7, 8, 11

9 pages

Review Result of the Programme Committee

This contribution has been conditionally accepted, pending any reviewers' comments being addressed.

Message from Chairs to Authors

This paper has been assigned to the track: CDIO Implementation

Review Results: Accept with minor changes.

The reviewers of your paper have highlighted the need for some minor revisions. Please repay their efforts by considering their suggestions when you revise your paper.

The reviewers comments are attached below.

Overview of Reviews

Questions		<u>Review 1</u>	<u>Review 2</u>	<u>Review 3</u>
Relevance	20%	5	5	5
Literature	15%	4	3	4
Aim or problem	15%	5	4	5
Research approach	15%	4	3	3
Originality	15%	5	4	4
Conclusions	10%	5	4	3
Coherence and Clarity	10%	5	5	3
Total points (out of 5)		4.7	4.0	4.0

Review 1

Evaluation of the Contribution

Relevance	Literature	Aim or problem	Research approach	Originality	Conclusions	Coherence and Clarity	Total points (out of 5)
5 (20%)	4 (15%)	5 (15%)	4 (15%)	5 (15%)	5 (10%)	5 (10%)	4.7

Reviewer's Comments on the Contribution

Manuscript formatting: Yes

Acceptance recommendation: Accept with changes

Track recommendation: Advances in CDIO

Contribution of the Submission:

The submission's primary contribution lies in proposing and articulating an assessment architecture that positions pull requests (PRs) as core assessment infrastructure in CDIO-aligned project-based courses. Rather than treating GitHub as a technical tool, the paper conceptualizes PRs as pedagogical artifacts that integrate technical change with observable evidence of collaboration, reasoning, responsiveness to feedback, documentation, and handover. The work contributes a coherent design rationale, demonstrates its implementation across two intentionally sequenced courses, and introduces the notion of feedback carry-forward within a persistent repository to support cumulative, rather than fragmented, evaluation. Additionally, the paper distills transferable design principles that extend beyond GitHub-specific contexts, offering practical guidance for making collaborative learning processes visible, attributable, and assessable in team-based project environments

Comments for the Authors:

This is a strong, well-written, and highly relevant contribution addressing a persistent challenge in CDIO-aligned project-

based education: how to make collaboration, accountability, responsiveness to feedback, and reproducibility visible and assessable. The manuscript is clearly structured, conceptually coherent, and grounded in authentic teaching practice. The reframing of pull requests (PRs) as assessment infrastructure rather than as a technical tool is particularly insightful and represents a meaningful contribution to discussions on assessment design in project-based learning contexts.

To further strengthen the paper, I offer the following constructive suggestions:

1. Strengthen the assessment-theory positioning

The literature grounding is appropriate and well integrated; however, the scholarly framing could be enhanced by deeper engagement with assessment-focused research. In particular, connecting the work more explicitly to literature on formative assessment, peer assessment, process-oriented evaluation, feedback literacy, or collaborative learning assessment would strengthen the theoretical contribution. This would help situate the proposed architecture within broader assessment discourse beyond GitHub-based contexts.

2. Clarify the evidence base

The practitioner/design-oriented framing is entirely appropriate for a CDIO practice paper. Nevertheless, including lightweight descriptive evidence would improve the paper's analytical depth. For example, indicative metrics (e.g., changes in PR size, review frequency, documentation quality), structured observations, or brief illustrative examples of PR evolution could enhance credibility without requiring a full empirical study.

3. Elaborate limitations and boundary conditions

The manuscript appropriately acknowledges its experience-based nature. This discussion could be expanded slightly to address issues such as scalability (large cohorts), disciplinary transferability (non-technical domains), variability in student readiness, and potential tensions between workflow overhead and learning benefits. Such reflection would increase the paper's value for diverse CDIO contexts.

4. Provide additional operational clarity

The transferable design principles are one of the paper's strongest elements. Their practical utility could be further enhanced by briefly clarifying assessment calibration mechanisms. For instance, examples or criteria illustrating what constitutes a "substantive peer review," or how review quality is evaluated, would improve reproducibility for other educators.

Review 2

Evaluation of the Contribution

Relevance 5 (20%)	Literature 3 (15%)	Aim or problem 4 (15%)	Research approach 3 (15%)	Originality 4 (15%)	Conclusions 4 (10%)	Coherence and Clarity 5 (10%)	Total points (out of 5) 4.0
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Reviewer's Comments on the Contribution

Manuscript formatting: Yes

Acceptance recommendation: Accept with changes

Track recommendation: CDIO Implementation

Contribution of the Submission:

This paper describes the use of pull requests (PR) for continuous assessment in industrial engineering courses. The paper discusses the pros and cons of different methods and how PR in GitHub can contribute to the assessment framework in open end group project courses.

Comments for the Authors:

The paper presents an interesting approach of assessment method in an open end project course. The paper is very well written, with a good flow and easy to read.

The paper is good as it is, but I would like to suggest the following changes that may improve the paper:

- The concept of pull requests (PR) is maybe not generally known to all CDIO community, it would be appreciated if a general description of this method would be introduced in the beginning of the paper
- Page 2: It is not clear if Information Engineering course is 2nd year undergrad or grad course
- How many students are usually enrolled into the courses, how many groups are formed?
- Is the literature review sufficient? A number of studies are presented, but if this is a well known method there might be more case studies to review and discuss
- Are there any quantifiable measures on evaluating this method over previously used methods? student evaluation? student performance in the course/other continuing courses?
- What are the main limitations of using this method over other/previously used ?

Good paper and well written, an interesting read, even for someone who is not an expert in this field

Review 3

Evaluation of the Contribution

Relevance 5 (20%)	Literature 4 (15%)	Aim or problem 5 (15%)	Research approach 3 (15%)	Originality 4 (15%)	Conclusions 3 (10%)	Coherence and Clarity 3 (10%)	Total points (out of 5) 4.0
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Reviewer's Comments on the Contribution

Manuscript formatting: Yes

Acceptance recommendation: Accept with changes

Track recommendation: CDIO Implementation

Contribution of the Submission:

The main contribution of this submission is the development of an assessment architecture that utilizes Pull Requests (PRs) as the primary unit of evaluation to make collaborative processes visible and assessable. Key Pillars of the Contribution: shifting the role of PRs from a technical tool to a framework that captures evidence of reasoning, peer review, and responsiveness to feedback in real-time; demonstrating a two-stage implementation where tool fluency is established in a foundational second-year course, allowing advanced courses to focus on high-quality collaboration and analytical work; introducing a "carry-forward" mechanism within a single persistent repository, ensuring that instructor feedback remains binding and actionable across multiple project cycles; providing a structural safeguard against "free-riding" by making individual contributions, critique quality, and documented follow-up traceable and attributable to specific students.

Comments for the Authors:

By establishing Pull Requests (PRs) as assessment infrastructure, the paper introduces a pedagogical innovation that is in line with CDIO Standards 7, 8, and 11. One of the highlights is the deliberate order of the third-year BI course and the second-year IE course. However, the paper needs more visual clarity regarding the interaction model and does not currently have empirical evidence to support its efficacy.

- The main issue is that the paper presents practitioner experience without reliable data or controlled experimentation. The author should include some quantitative data, collaboration metrics, and/or student perception to support the "Learning Assessment" (Standard 11) claim.
- The paper describes complex workflows but lacks a visual representation of the interaction between stakeholders. This should be improved in the final version.
- The author claims the model fosters "professional competence" and "shared responsibility". However, the link between specific CDIO skills and measurement methods should be more explicit.